B. Tech C.S.E. 8th Semester Syllabus

PAPER CODE	Subject	CREDITS (T+P)
100708	Bitcoin and Crypto Currencies	3
100713	Cloud Computing	3
105707	Embedded Systems	3
105718	Ad-hoc and Sensor Networks	3
100709P	Project – II	6

Bitcoin and Crypto Currencies

PAPER CODE-100816

Module 1 hrs. : Lecture 8

Introduction to Cryptography, Cryptographic Hash Functions, SHA-256, Hash Pointers and Data Structures, Merkle tree.

Module 2 : Lecture 8 hrs.

Digital Signatures, Elliptic curve group, Elliptic Curve Digital Signature Algorithm (ECDSA). Public Keys as Identities, A Simple Crypto currency.

Module 3 : Lecture 8 hrs.

Centralization vs. Decentralization, Distributed consensus, Consensus without identity using a block chain, Incentives and proof of work. Bitcoin transactions, Bitcoin Scripts, Applications of Bitcoin scripts, Bitcoin blocks, The Bitcoin network.

Module 4 : Lecture 8 hrs.

Simple Local Storage, Hot and Cold Storage, Splitting and Sharing Keys, Online Wallets and Exchanges, Payment Services, Transaction Fees, Currency Exchange Markets.

Module 5 : Lecture 8 hrs.

Bitcoin Mining, Mining pools, Mining incentives and strategies. Bitcoin and Anonymity: Anonymity Basics, Mixing, Zerocoin and Zerocash.

Cloud Computing

PAPER CODE- 100817

Module 1 : Lecture 4 hrs.

Introduction: Distributed Computing and Enabling Technologies, Cloud Fundamentals: Cloud Definition, Evolution, Architecture, Applications, deployment models, and service models.

Module 2 : Lecture 5 hrs. :Virtualization

Issues with virtualization, virtualization technologies and architectures, Internals of virtual machine monitors/hypervisors, virtualization of data centers, and Issues with Multi-tenancy.

Module 3 : Lecture 6 hrs. : Implementation:

Study of Cloud computing Systems like Amazon EC2 and S3, Google App Engine, and Microsoft Azure, Build Private/Hybrid Cloud using open-source tools, SLA management.

Module 4 : Lecture 12 hrs. : Resource Management:

Cloud resource provisioning plan (advance reservation, on demand plan, spot instances), various scheduling and load balancing techniques to improve QoS parameters, Resource Optimization algorithms, task migration and VM migration technique.

Module 5 : Lecture 7 hrs. : Security:

Vulnerability Issues and Security Threats, Application-level Security, Data level Security, and Virtual Machine level Security, Infrastructure Security, and Multi-tenancy Issues.

Module 6 : Lecture 6 hrs. : Advances:

Green Cloud, Mobile Cloud Computing, Fog Computing, Internet of Things

Embedded Systems
PAPER CODE-105816

Module 1 : Lecture 10 hrs.

Embedded Computing: Introduction, Complex systems and

Microprocessors, The embedded system design process, Formalization for system design.

Module 2 : Lecture 10 hrs.

Instruction Sets CPUs: Instruction and preliminaries ARM and

SHARC Processors, Programming I/O CPU performance and Power consumption.

Module 3 : Lecture 10 hrs.

The embedded Computing Platform and program design: Introduction, the CPU bus, Component interfacing, designing with microprocessors, development and debugging.

Module 4 : Lecture 10 hrs.

Program Design and Analysis: Introduction program design,

Assembly, Linking, Basic compilation techniques, and Analysis optimization of executive time.

Ad-hoc and Sensor Networks
PAPER CODE- 105819

Module 1: <u>Introduction Lectures 8hrs.</u>
Fundamentals of wireless communication technology - the electromagnetic spectrum - radio propagation mechanisms - characteristics of the wireless channel - Mobile Ad-hoc Networks (MANETS) and Wireless Sensor Networks (WSNs): concepts and architectures. Applications of Ad-hoc and sensor networks. Design challenges in Ad-hoc and sensor networks.

Module 2: <u>Mac Protocols for Ad-hoc Wireless Networks Lectures 8 hrs.</u>
Issues in designing a MAC Protocol- Classification of MAC Protocols- Contention based protocols Contention based protocols with Reservation Mechanisms- Contention based protocols with Scheduling Mechanisms - Multi channel MAC-IEEE 802.11

Module 3: <u>Routing Protocols and Transport Layer in Ad-hoc Networks Lectures 8 hrs.</u>
Issues in designing a routing and Transport Layer protocol for Ad hoc networks- proactive routing, reactive routing (on-demand), hybrid routing- Classification of Transport Layer solutions-TCP over Ad hoc wireless Networks.

Module 4: <u>Wireless Sensor Networks (WSNs) And MAC Protocols Lectures 8 hrs</u>.
Single node architecture: hardware and software components of a sensor node - WSN Network architecture: typical network architectures-data relaying and aggregation strategies -MAC layer protocols: self-organizing, Hybrid TDMA/FDMA and CSMA based MAC- IEEE 802.15.4

Module 5: <u>Security in Ad Hoc and Sensor Networks Lectures 8 hrs.</u>
Security Attacks - Key Distribution and Management -Intrusion Detection - Software based Anti-tamper techniques - Water marking techniques - Defense against routing attacks Secure Ad hoc routing protocols - Broadcast authentication WSN protocols - TESLA - Biba - Sensor Network Security Protocols - SPINS